

Xinyi (Jesse) Yang

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PROFILE

M.Sc. Computer Science (AR/VR) student at Trinity College Dublin with a background in game design and experience in Unity, real-time graphics and shader development. Interested in graduate Technical Art and Graphics roles, particularly materials, shaders and real-time rendering.

EDUCATION

Trinity College Dublin, Dublin, Ireland

Sep 2026 – Aug 2027 (Expected)

M.Sc. Computer Science - Augmented and Virtual Reality

- Modules include: Advanced Software Engineering, Machine Learning, **Computer Graphics**, Computer Vision, **Real-Time Rendering**, **Real-Time Animation**

University of Wisconsin–Madison, Madison, WI, USA

Sep 2021 – May 2025

B.S. Computer Science, Certificate in Game Design | GPA: 3.936 / 4.0

- Dean's List all terms; Distinctive Scholastic Achievement

PROJECTS

Real-Time Material and Shader Projects (Unity URP, HLSL, ShaderLab) Aug 2026 - Oct 2026

- Built 2 real-time shader projects in Unity URP using HLSL/ShaderLab, with modular shader code and exposed Material Inspector controls for rapid visual iteration.
- Developed a configurable three-tone toon shader with adjustable lighting thresholds and colors, optional texture mapping, color tinting, and view-dependent rim lighting for stylized character rendering.
- Created a procedural weathered-metal shader using multi-octave FBM noise and edge-wear masking to generate localized rust and surface damage, procedurally varying albedo, metallic and smoothness under URP PBR lighting; prepared UV-unwrapped test geometry in Blender.

Computer Graphics and Rendering (C++, OpenGL, GLSL, Eigen)

May 2026 - July 2026

- Implemented an OpenGL pipeline from VAO/VBO/EBO setup and GPU buffer uploads through GLSL shader compilation and indexed/non-indexed rendering.
- Implemented rasterization and shading in C++/Eigen, including Z-buffering, SSAA, barycentric interpolation, Blinn-Phong shading, texture/bump/displacement mapping, and ray/path tracing with BVH (SAH), microfacet BSDF and Multiple Importance Sampling.
- Built Bézier curves and mass-spring cloth/rope simulations using de Casteljau and explicit/semi-implicit Euler integration.

Gameplay Programmer and Co-founder, Old Man & Dog Studio (remote)

Feb 2026 – Present

- Program gameplay for a narrative RPG in Godot 4 as part of a five-person indie team.

RELEVANT EXPERIENCE

Unity Gameplay Development Intern, Merfolk Games, Shanghai, China

Jun – Aug 2026

- Shipped object-oriented C# gameplay modules in an existing production Unity codebase for a live interactive exhibit at the Shanghai Museum, using Git-based team workflows.
- Debugged and refactored Unity Animator state machines to resolve transition/blending defects, and implemented event-driven audio plus purchase/reward subsystems driven by game state.
- Added Unity Test Framework coverage and participated in pull-request review; one test caught a defect before it reached the build.

Undergraduate Researcher, NSF Athena REU, WINGS Lab, UW–Madison Jun – Sep 2024

- Designed and implemented 3D volumetric mapping and real-time navigation in Unity/C# for AR smart glasses, supporting first-responder research.
- Integrated PyTorch object recognition (~92% classification accuracy) and converted backend detections into live AR overlays.
- Built event-driven C# integration for streaming JSON sensor data and Unity NavMesh routing across obstacles, doors and stairwells in complex 3D environments.

OTHER TECHNICAL EXPERIENCE

Full Stack Developer, Kanger Technology Co., Ltd (remote) Feb – Jul 2025

- Developed and maintained a production Next.js/React platform with Prisma and PostgreSQL serving 3,000+ monthly users, with NextAuth authentication and role-based access control.
- Built tiered pricing, real-time inventory synchronization (≤ 100 ms), referral tracking and automated sales commissions; migrated deployment from Vercel to Alibaba Cloud using Nginx and PM2, cutting hosting cost by 30% and time-to-first-byte by 40%.

Research Assistant, Social Interaction Lab, Wisconsin Institute for Discovery May – Aug 2024

- Built and parallelized a Python audio-processing pipeline using OpenAI Whisper across 50+ hours of recordings, improving throughput by ~50% through multiprocessing.
- Automated NLP classification using the OpenAI API, reducing manual processing time by ~90% at ~95% accuracy, and analyzed results in Pandas.

TECHNICAL SKILLS

Programming & Shaders: C++, C#, Python, HLSL, GLSL, JavaScript/TypeScript, SQL

Graphics & Engines: Unity (URP, ShaderLab, Animator, Unity Test Framework), OpenGL, Blender, Godot 4; PBR/toon shading, procedural noise, rasterization, ray tracing, path tracing

Tools & Software Engineering: Git, CMake, Docker, Eigen, object-oriented programming, unit testing, code review, debugging, event-driven systems, data structures and algorithms

Computer Vision & Data: PyTorch, OpenCV, Pandas, OpenAI Whisper, multiprocessing

Web & Backend: React, Next.js, NextAuth, Prisma, PostgreSQL, Nginx, PM2, AWS